① The problem is very technical and social aims are unclear.
② What is the niche? I do not see a discourse here.

③ Literature is missing

Stock Price Forecasting Using Natural Language Processing and Market Movement Analysis

REWRITE

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Abstract

citations

Existing research on stock movement prediction, often relying on multiple datasets and code implementations, typically shows **low accuracy**, usually around **50%**. Only a few studies achieve accuracies of **55% to 60%**. These projects commonly use only the **yfinance dataset** and are usually limited to predicting the **direction of the next day's price movement** (up or down), without providing a forecast for the **magnitude** of the movement.

Big Shot?

We acknowledge that stock movements are highly dependent on **unknown variables**, and 100% accurate prediction is impossible. If such a capability existed, stakeholders with this knowledge would have an insurmountable edge, leading to effectively infinite gains and likely **destabilizing the entire global market**. Similarly, if this perfect prediction were the secret of a few wealthy individuals, the resulting excessively high gains would discourage others from investing in shares or financial assets.

Isn't it already done? AI use is huge in finance

However, we believe that combining **recent stock news** with **historical stock prices** can significantly improve prediction accuracy. Even a moderate improvement can already aid decision-making for buying, selling, or choosing positions in the stock market.

Therefore, our goal is to **collect all possible information** related to a specific stock, such as **Apple**, including news, historical data, and target prices. We will use this comprehensive dataset to train a machine learning model to predict stock movements with a **fair accuracy**, hopefully reaching around **60%**. This level of accuracy, while not perfect, should be **sufficient to support human decisions** on options and orders in the stock market, ultimately helping stakeholders achieve higher gains.

Data

The **data** is, of course, the most crucial component: what data to use, how to collect it, and how to process it to ensure it is **statistically significant and representative**.

We have two main data categories:

1. **Stock News and Fetchable Internet News**: Any news with an exact publishing date and information relevant to a specific stock.
2. **Historical Stock Prices and Volume**: Data from reliable sources.

Initially, to make the task manageable, we will **reduce the scope** by focusing on a **single large-cap stock** like **Alphabet, Amazon, or Apple.**

Collecting news data is complex. While some related stock news datasets are available on Kaggle (for example, for Google), most are **not up-to-date** and cover only a **short time range**. This is not necessarily a huge problem, as we may be able to **train the model** with a short-range, older dataset and use **more recent news as a test set** to check accuracy. If the news types are similar and the training and test sets are balanced, the model may still perform well. However, this immediately presents the challenge of the model using **old stock prices** to adjust predictions based on new news, while the stock prices have significantly changed since the dataset's end date.

Since we plan to use a **supervised training method**, we must select **target variables**. The convenient **Yahoo Finance (yfinance) library** allows us to easily fetch valuable and up-to-date stock information into a Python environment. The goal is to predict price movement with maximum accuracy, so **open, daily high, daily low, close prices, and volume** appear to be useful target variables.

Stakeholders

While **government bonds** are sometimes a secure alternative for preserving capital's buying power, they do not consistently guarantee a profit. **Portfolio diversification** is critical, but the aim is to **maximize profit** while staying within an acceptable risk range. Although governmental bonds are secure and often have low tax obligations, there have been years where high inflation eroded buying power (though inflation-indexed bonds exist).

In recent years, people have sought alternatives, and multiple broker companies have targeted European and UK citizens, offering easy and convenient platforms to buy financial assets. For instance, Interactive Brokers introduced a special tax-advantaged account (TBSZ) for Hungarian citizens, a service previously limited to local banks.

The easier, cheaper, and more popular access to buying shares and other financial assets has led to an increase in **small investors** buying large-cap stocks like Microsoft, Amazon, and Nvidia. These small investors are key **stakeholders** for the decision-support service we are researching, as they often **lack immediate access to information** from closed professional circles or top-tier decision-helping software.

Methodology

Our objective is to not only analyze the news for **price movement direction** but also to **estimate the magnitude (volume)** of that movement. To achieve this, we must **categorize stock news** to increase model accuracy and decrease loss. Simple sentiment analysis, as used in most similar research, only yields a score for the article, which may help predict direction but not volume. Therefore, we need to **differentiate major market events** and identify the primary constraints on stock movements.

citations

This requires collecting several **dynamic variables** that could affect a specific share price and **categorizing the news** accordingly. For example, a company's stellar quarterly report might generate very positive sentiment, but a much stronger effect on future expectations and the stock price would come from an announced **acquisition of another company of comparable size**. The latter event could significantly increase the stock's volatility and potential gain.

Potential categories for news differentiation include:

- **Firm Aspects**: Company reports, product launches, management changes.
- **Macroeconomic Changes**: Interest rates, inflation, GDP, exchange rates.
- **Psychological Factors**: Investment sentiment, market mood.
- **Technical Factors**: Trading volume, technical trends.

We will categorize the news into these groups, as the category itself is an **important predictive factor**, not just the sentiment. Alternatively, we could build a **high-dimensional space** to store keywords and use models like **Naive Bayes** or others for more accurate news differentiation. We plan to use a **supervised training method** and will test multiple models, including **Naive Bayes, Logistic Regression**, and others, to find the best fit.

Methodology

Our objective is to predict not only the direction of price movements from news, but also the magnitude of those moves. To do this, we go beyond simple sentiment scores and categorize news by event type, which improves signal quality and reduces model loss. Basic sentiment alone often indicates *up vs. down* but rarely explains *how much* prices move. Differentiating major market events and the constraints they impose on price dynamics is therefore essential.

We collect multiple dynamic variables that influence a specific stock and align them to a common timeline. For example, a strong quarterly report may create positive sentiment, but an M&A announcement of comparable scale typically shifts expectations far more and increases both volatility and potential returns. Capturing such differences helps the model learn when large moves are more likely.

We organize news into four practical categories:

- Firm Aspects: earnings, guidance, product launches, leadership changes, M&A.
- Macroeconomic Changes: interest rates, inflation, GDP releases, FX shocks.
- Psychological Factors: investor sentiment, risk appetite, crowd mood.
- Technical Factors: unusual volume, momentum/trend breaks, liquidity shifts.

Both category and sentiment are used as predictors alongside historical OHLCV features. For text differentiation, we construct a high-dimensional keyword space and test lightweight classifiers (e.g., Naïve Bayes) for event tagging. The forecasting task is framed as supervised learning with strict time ordering. We evaluate several models—Naïve Bayes, Logistic

Regression, and tree ensembles—choosing the best fit based on out-of-sample accuracy for direction and error (e.g., MAE) for magnitude.

Research Paper References

1.

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- **Author:** Cenk R. Dogan (crndogan)
- **Year:** 2023 (Based on repository activity/creation)
- **Title:** stock-news-prediction (Specifically referencing the metrics.csv file/dataset)
- **Type/Source:** GitHub (Data set/Source code)
- **URL:** https://github.com/crndogan/stock-news-prediction/blob/main/notebooks/metrics.csv

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This appears to be a personal blog post or project page hosted on GitHub Pages.

- **Author:** Jin Choi (choijin)
- **Year:** 2023 (Estimate based on content activity/creation)
- **Title:** NLP Text Stock Prediction
- **Type/Source:** Blog post / Website
- **URL:** https://choijin.github.io/NLP_Text_Stock_Prediction/

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This is an article from the educational website GeeksforGeeks.

- **Author:** The article typically credits the specific author or uses **GeeksforGeeks** as the organizational author. (Checking the page would be necessary for the specific author, but **GeeksforGeeks** is a safe default.)
- **Year:** The article's most recent update or publication year (Please check the page for a specific date). *Using **2023** as an example.*
- **Title:** Stock Price Prediction Using Machine Learning in Python
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